*hala tuju = direction*

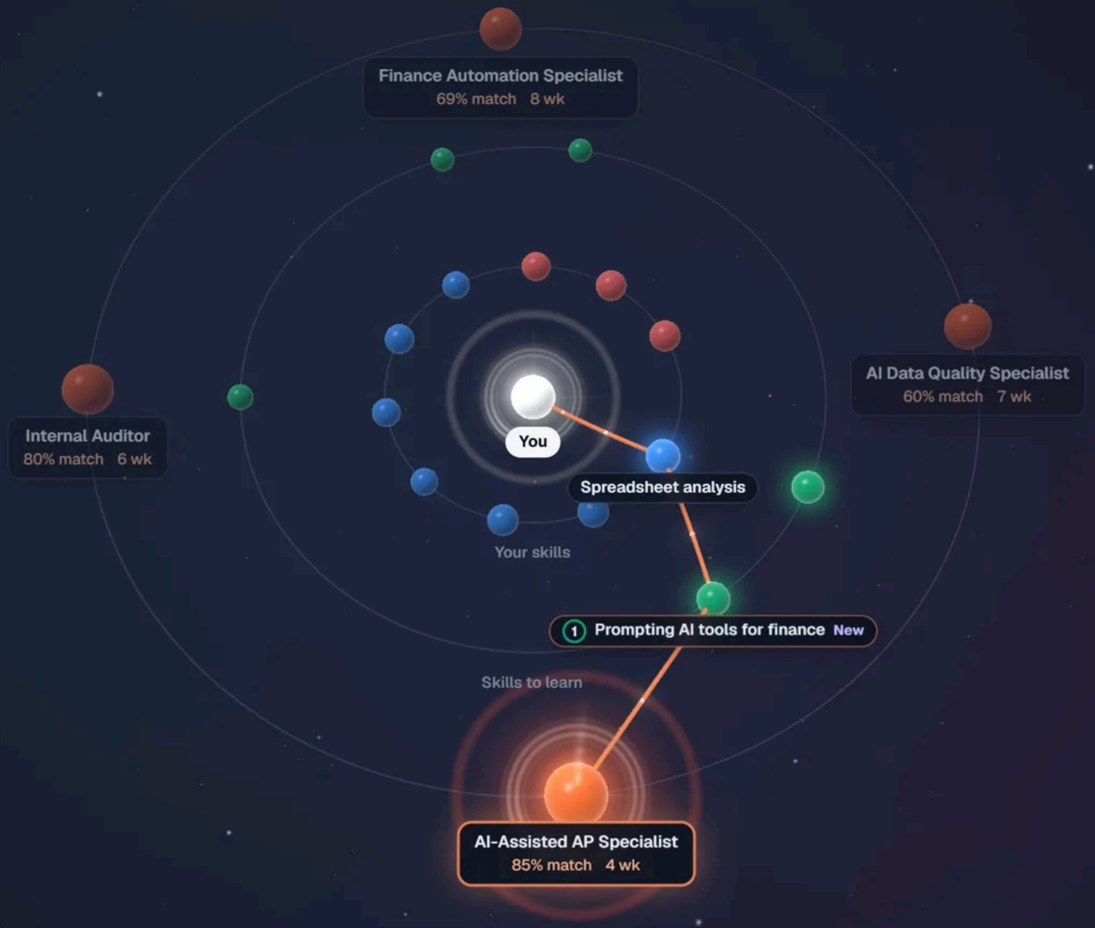
# HalaTuju

## A GPS for your career: when AI closes a road, it reroutes you.

SDC Hackathon 2026, Domain 1: Aiversecity Future-Ready Workforce


**TEAM DSAISE**

Leow Shen En
Lim Jie Shin
Choong Zhuo Lin
Tan Wei Feng
Esther Kong Yuan Er

*The problem*

# AI is reshaping jobs one task at a time

**620k**

Malaysian jobs highly impacted by AI and digitalisation within 3–5 years

TalentCorp impact study, 2024

**1 in 4**

workers worldwide have a job that AI can partly do. Most of these jobs will change, not vanish.

ILO Working Paper 140, 2025

**57%**

of AI use at work helps a person do the task. The other 43% does it alone.

Anthropic Economic Index, 2025

“

**For many jobs, AI can only do 30–40 percent of the work now and for the foreseeable future. That means we still need people to do the remaining 60–70 percent.**

Andrew Ng, World Economic Forum, Davos, January 2026

> HalaTuju shows each worker their 60–70%: the skills AI can’t do, and the jobs those skills lead to.

*The gap*

# Today's tools match skills, not routes

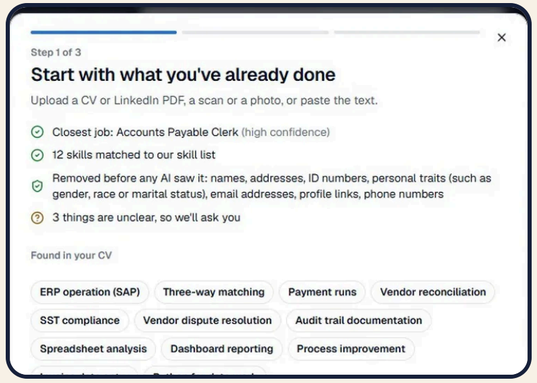
| | Skills matching | AI exposure | Step-by-step plan with dates | Updates when AI changes |
|---|---|---|---|---|
| **MyMahir**<br>TalentCorp | Skills test and role comparison | Job-level impact study | Upskilling suggestions | *Not stated* |
| **MYFutureJobs**<br>PERKESO | AI matching to vacancies | *Not stated* | Reskilling programmes | *Not stated* |
| **Career Explorer**<br>LinkedIn | Skills similarity score | *Not stated* | Skill gaps and courses | *Not stated* |
| **HalaTuju**<br>this project | ✓ 13,939 skills, 3,039 jobs | ✓ Per task, from 4 studies | ✓ Week by week, with costs | ✓ Daily, checked by a person |

Sources: Bernama 2024, PERKESO 2025–26, LinkedIn 2020. "Not stated" means not in public material as of Oct 2026.

*The product*

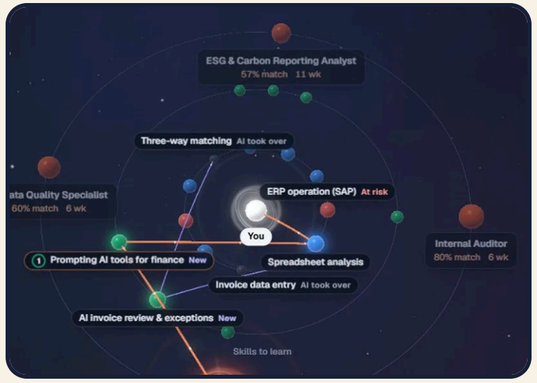
# How HalaTuju helps a worker

## 1 Upload your CV



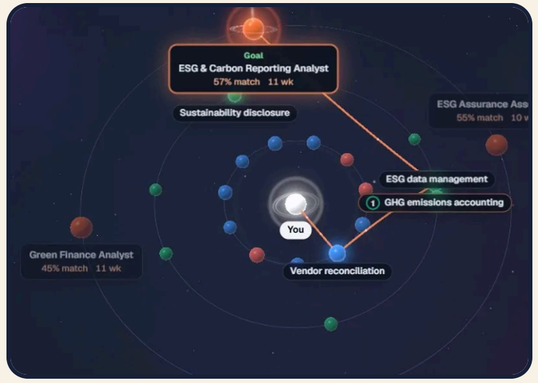

Your private details are removed first. Your skills appear in seconds.

## 2 See your skills



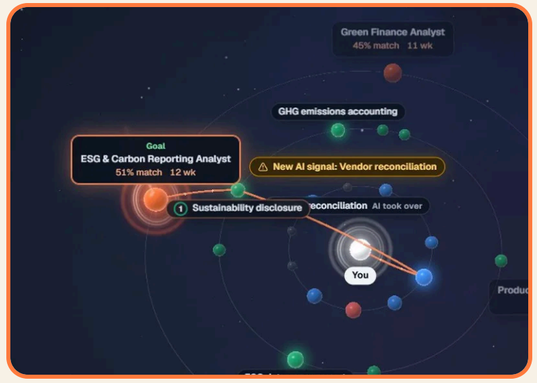

See which skills will still count, and which AI is taking over.

## 3 Pick a route




Choose a new job and get a study plan, with time, cost and courses.

## 4 Stay on track

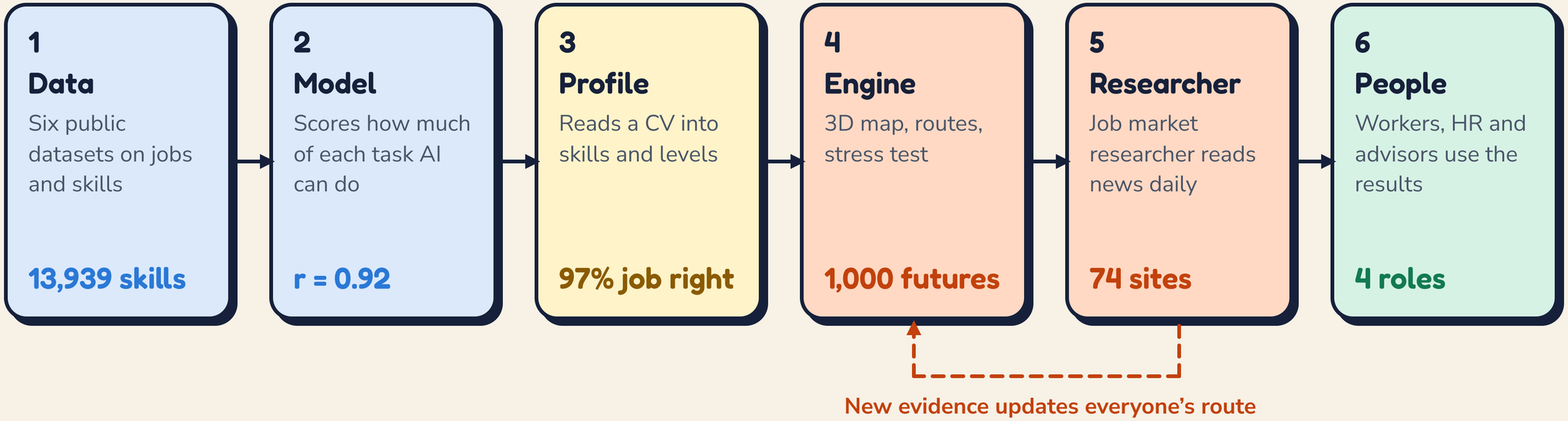



Our job market data stays up to date, and your plan adjusts as it changes.

Screens from the live app at halatuju.replit.app.

*The system*

# How it’s built: six layers, from data to people




Runs on Replit with TypeScript and Postgres. Gemini Flash-Lite reads, bge-m3 on RunPod matches, and 419 automated tests check it.

# Built on six public datasets

**13,939**
skills across 3,039 jobs

**ESCO v1.2.1, European Commission**

**16,411**
tasks on ESCO-linked jobs, rated by experts for AI

**O*NET with GPTs-are-GPTs labels**

**2,956**
jobs scored for generative-AI exposure

**ILO Working Paper 140, 2025**

**Real use**
how people actually use AI at work

**Anthropic Economic Index**

**2021–25**
wages and employment, 9 job groups

**DOSM, via OpenDOSM**

**66**
critical occupation groups in Malaysia

**TalentCorp MyCOL 2024/25**

Plus 30 real course listings with price, hours and link. Every number in the app names its source.

# Our AI-exposure model agrees with experts

All 19,265 tasks rated by researchers → bge-m3 turns each task into numbers, on RunPod → A simple model learns the expert ratings → Scores all 13,939 skills

**0.92**

agreement with experts on 185 jobs it never saw

Better than GPT-4's ratings (0.85) and a simple baseline (0.88)

**0.79**

agreement with the ILO's index on 2,956 jobs

An independent check: we never trained on ILO data

**4**

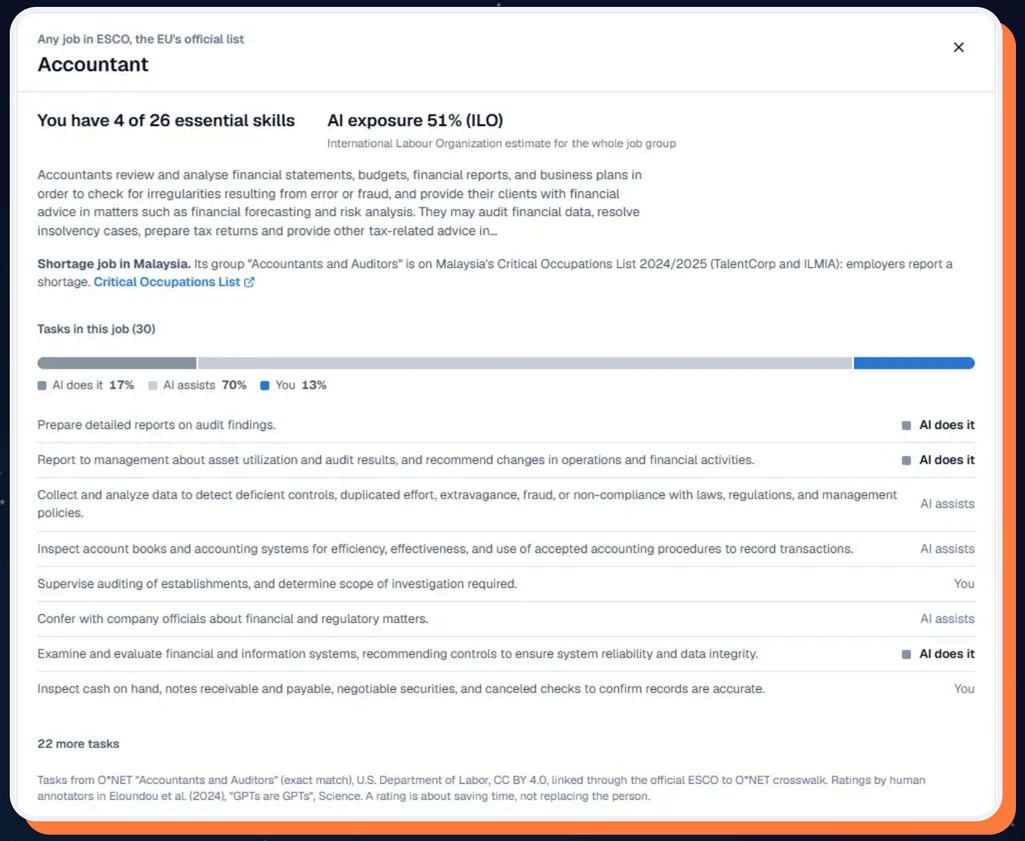
AI-exposure sources shown side by side

ILO, Anthropic, GPTs-are-GPTs and our model

Expert ratings from Eloundou et al., Science 2024. Test jobs were kept out of training. r is the correlation per job (1 = perfect).

1 Data | 2 Model | 3 Profile | 4 Engine | 5 Job market researcher | 6 People

# See how AI affects any of 3,039 jobs




**Your match**

You have 4 of the 26 essential skills

**AI exposure from four studies**

35–59% for accountants, from the ILO, Anthropic, GPTs-are-GPTs and our model

**Task split**

Of 30 tasks, AI does 17%, helps with 70%, and 13% still need a person

**Malaysian context**

On the critical-jobs list, RM 6,124 median pay, 9.8% more workers since 2021

Live screenshot. Sources: ESCO v1.2.1, O*NET via the official crosswalk, ILO WP140, DOSM 2025 and MyCOL 2024/25.

# From CV to skill profile in 15 seconds

**1 Upload**
CV or LinkedIn PDF, or pasted text

**2 Scrub**
Names, MyKad, phone, religion removed before any AI reads it

**3 Match**
Gemini reads it; bge-m3 matches ESCO skills

**4 Ask**
Up to 3 follow-up questions, EN or BM

**5 Level**
1–5 per skill: from CV, self-rated or quiz-verified

**97%**
of current jobs identified (35 of 36)

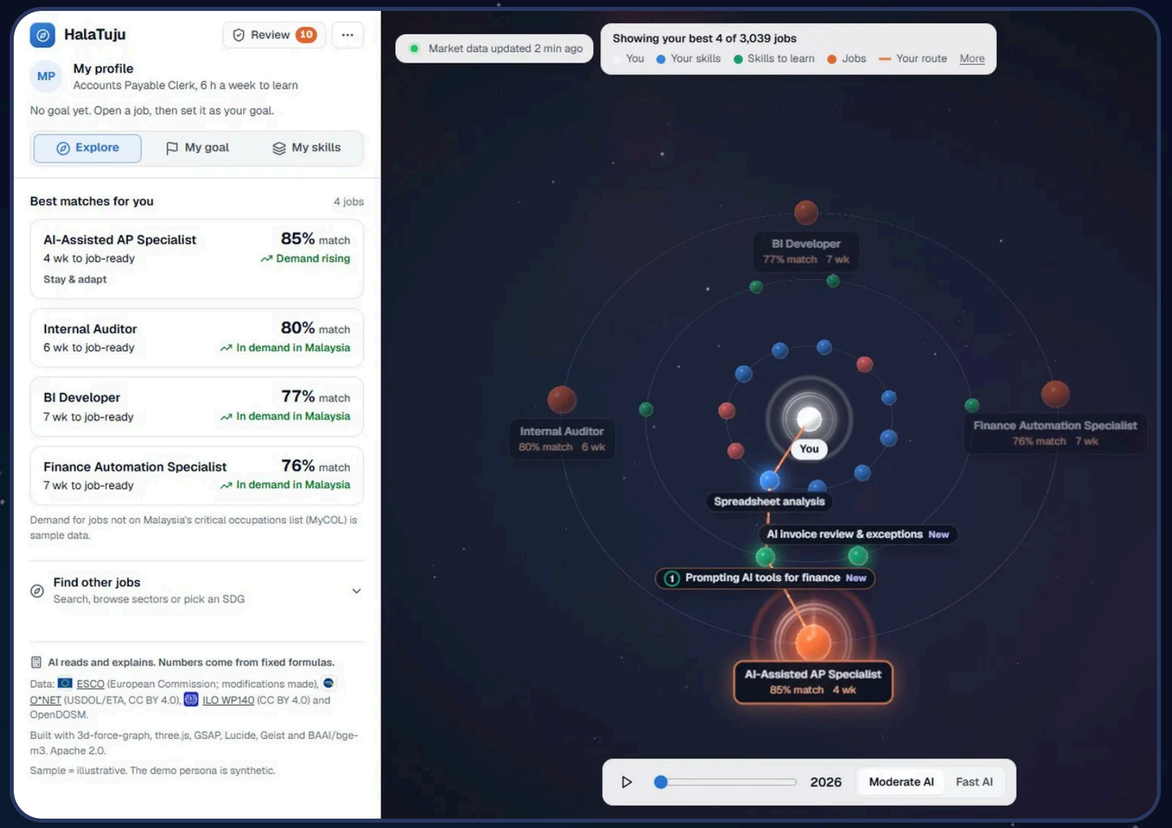
**94%**
of the expert's map skills found (84 of 89)

**15 s**
from upload to skill map, Priya's CV

**99.4%**
of personal details removed before the AI (154 of 155)

Blind answer key, 36 synthetic CVs we never tuned on (24 Oct 2026). Privacy 99.4% is after fixes; the blind first run was 82.6%.

# The 3D career map



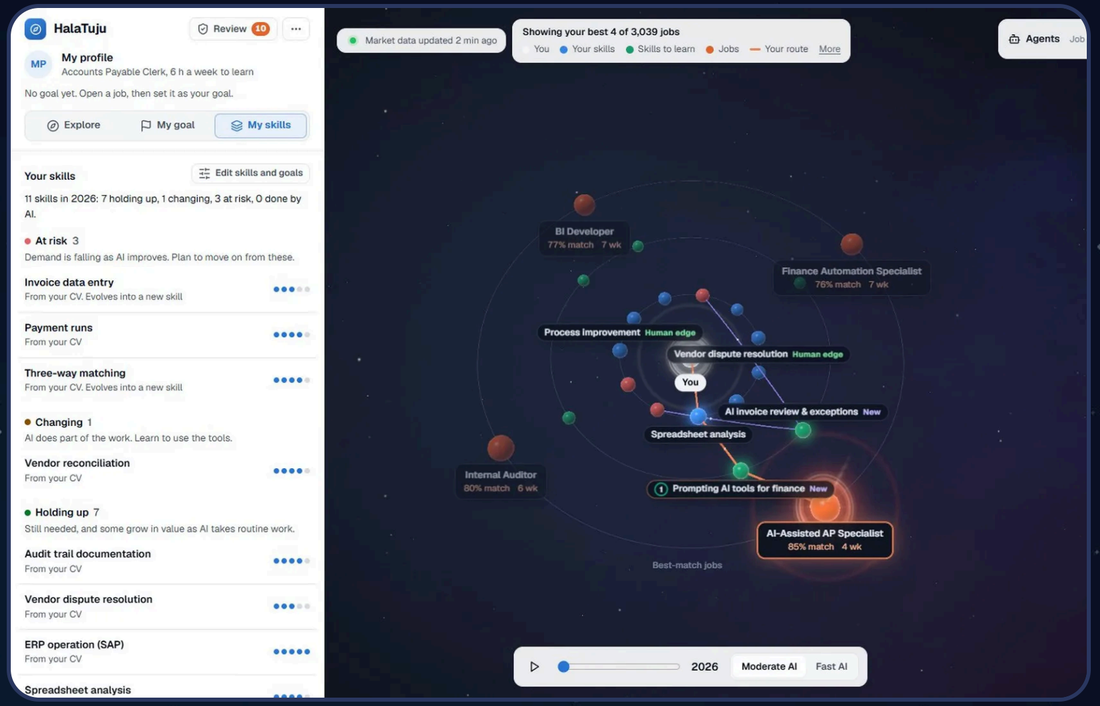

- **Centre: you**
- **Inner ring: your skills.** Red means AI already does it
- **Middle ring: skills to learn,** including new skills
- **Outer ring: jobs,** with match % and weeks
- **Orange line: your route,** step by step

Controls: year 2026–2030, steady or fast AI spread, sector and SDG focus. Click any node for its details.

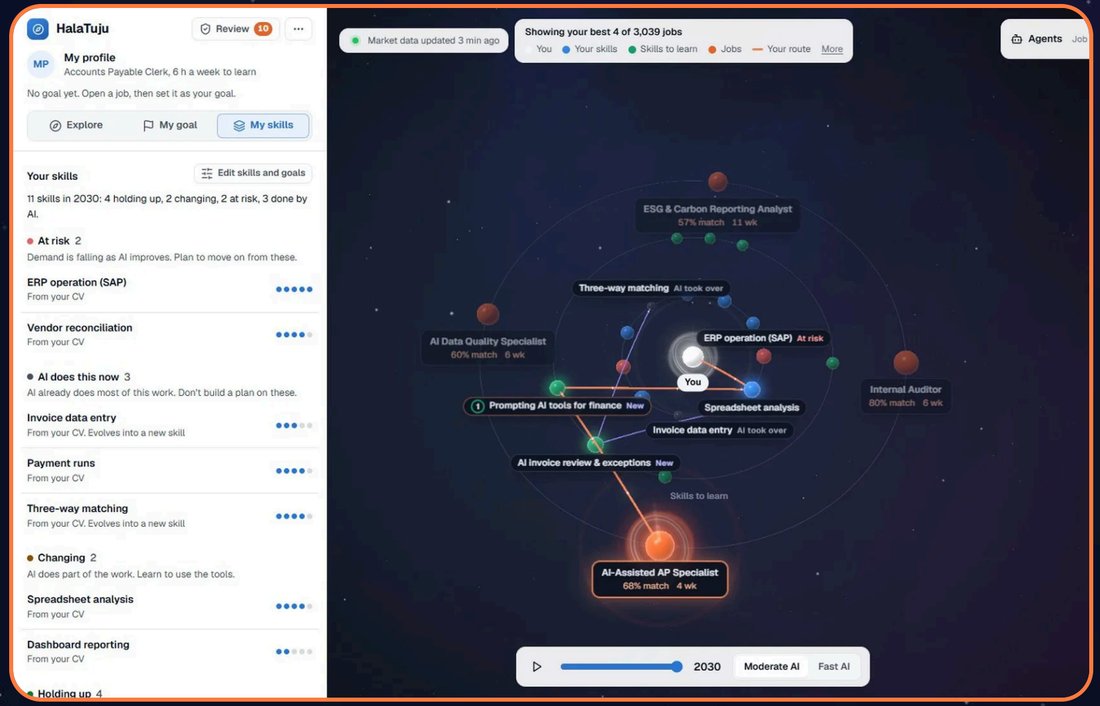
From the demo video: Priya, an Accounts Payable Clerk, and her best 4 of 3,039 jobs. Sample values are labelled in the app.

# A time machine: see your skills in 2030

## 2026: 7 of her 11 skills hold up

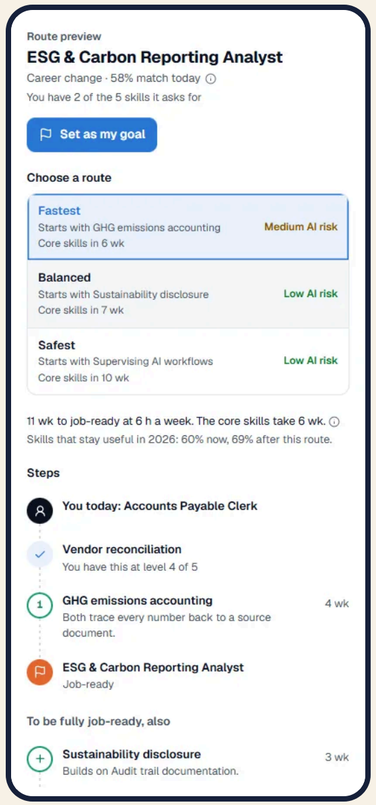



## 2030: AI does 3 of them; 4 hold up




From the demo video. Invoice entry goes to AI in 2027, three-way matching in 2028. Switch AI spread to Fast and it happens sooner.

# Three routes, priced in weeks and RM




**Fastest**
Core skills in 6 wk
Medium AI risk

**Balanced**
Core skills in 7 wk
Low AI risk

**Safest**
Core skills in 10 wk
Low AI risk

## How a route is found

Like a GPS, it searches every path from your skills to the job. Each step costs weeks of study at your pace, plus extra for skills AI may take. Safest avoids those most.

### Step types

Each step is reuse, upskill, evolve or reskill, done as learn, practice, prove

### Costed

Course fees in RM, and how much HRD Corp can cover

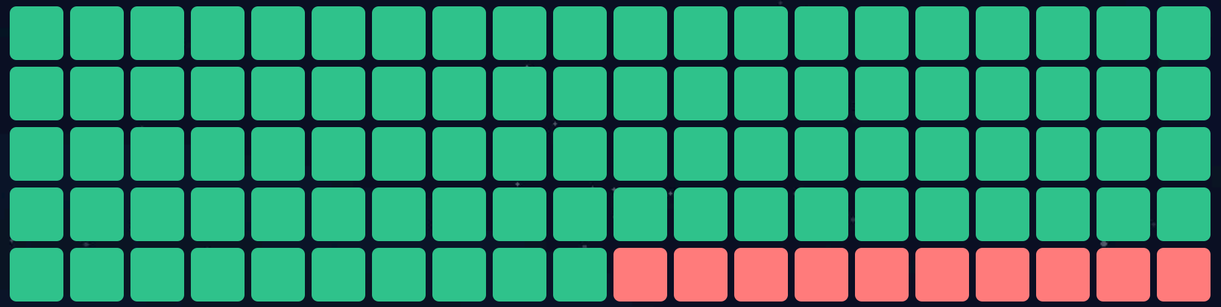
### Deadline

Shows if you are ahead, on track or behind your date

Example: AP clerk to ESG analyst in 11 weeks at 6 hours a week, with free courses.

# Every plan is tested in 1,000 futures

**Priya’s safest route: works in 90% of 1,000 futures**



Each square is 10 simulated futures. Red, about 1 in 10: not ready by the deadline. Safest steers around skills AI is taking, so it is the longest route and runs late when study drops below 4.5 hours a week.

## Random in each future

- AI speed, from slow to fast
- Our model’s measured error
- Study hours, half to 1.5 times the plan
- Each step’s length, ±20%

**Fastest** 99%, **Balanced** 100%, **Safest** 90%

The app suggests a fix: take Balanced (100%), or study 1 more hour a week (98%).

Seeded Monte Carlo: same plan, same number, within about ±2 points. It tests a plan; it does not predict hiring.

# An AI researcher checks 74 trusted sites daily

An AI agent that reads AI and job news every day, so the skill map and every route stay current.

**1 Collect**
RSS, arXiv, OpenAlex, GDELT, web search on vetted sites

→

**2 Extract**
Gemini finds a dated claim about a skill, with a quote

→

**3 Verify**
The quote must appear word for word on the page

→

**4 Review**
A second model checks the claim against its evidence

→

**5 Triage**
Small changes apply; the rest go to a person

**769**
articles in one run

**5.5 min**
per run

**28**
targeted searches

**30**
AI fact checks

**7**
flagged for a person

## WHAT IT LOOKS FOR

**New AI abilities**
AI can now do part of a skill's tasks

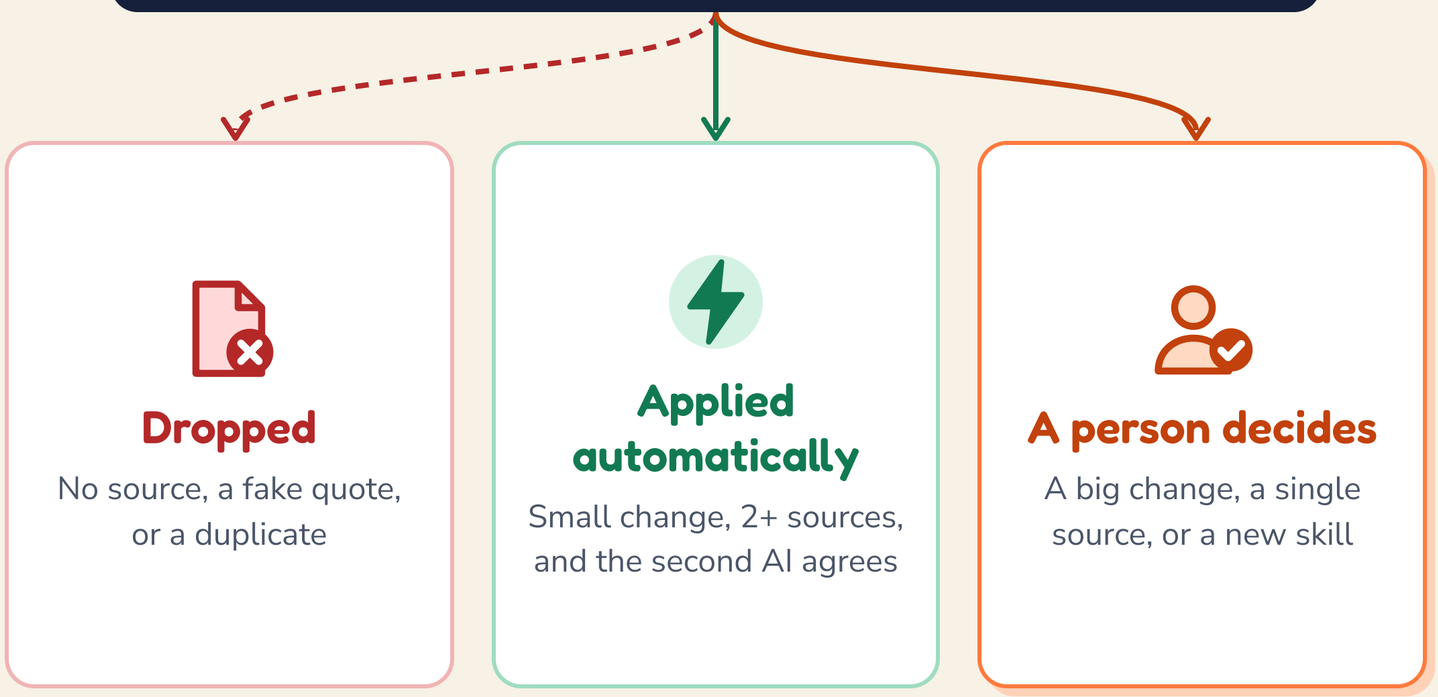
**Adoption**
Companies putting AI to work on those tasks

**Demand**
Employers needing more or fewer people with a skill

Live run, 23 Oct 2026. Each daily run covers 10 skills, so all 29 map skills every 3 days, plus 10 broad topics. Next: all 13,939.

# Triage: a person approves every big change

**Every finding is checked first**
Trusted source, quote found word for word, then a second AI reviews it




**Real signal, 22 Oct 2026**

*"...the Account Reconciliation Agent can help identify exceptions between subledgers and the general ledger and suggest how to resolve them..."*

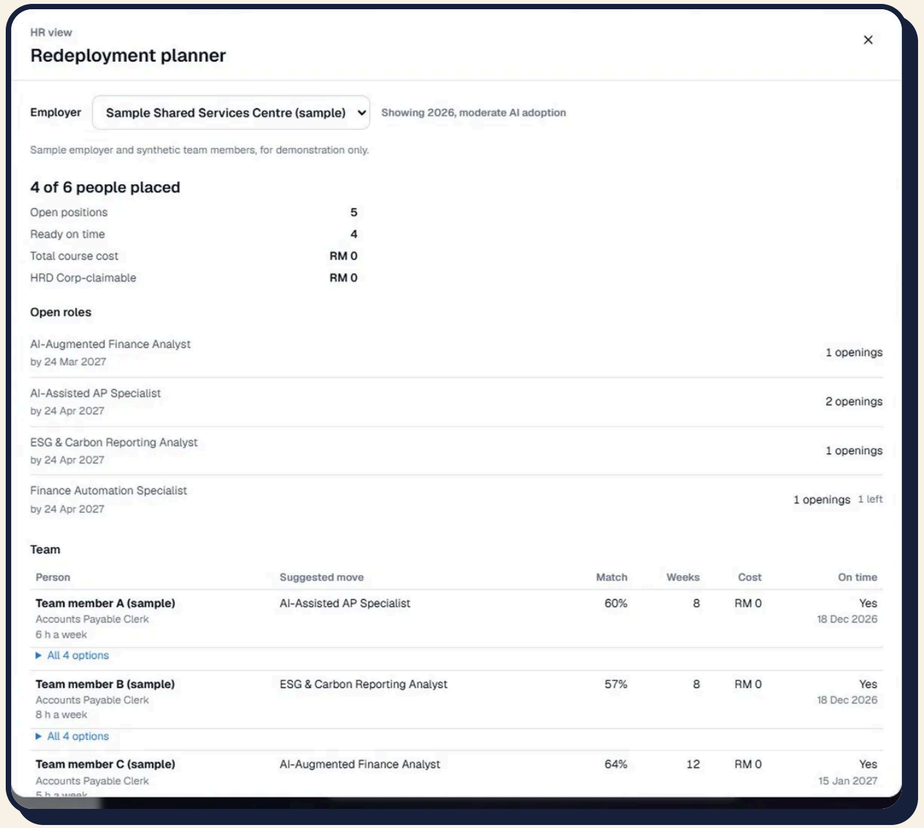
Microsoft Dynamics 365 blog, quote found on the page

**Vendor reconciliation:** AI exposure up, single source, so it went to a person

**Approved** → every route through reconciliation is re-planned

Every change is saved as a new version and can be undone. A new skill counts at half weight until a person confirms it.

# Employers and advisors use the same map



HR view

Redeployment planner

Employer: Sample Shared Services Centre (sample) — Showing 2026, moderate AI adoption

Sample employer and synthetic team members, for demonstration only.

**4 of 6 people placed**

| | |
|---|---|
| Open positions | 5 |
| Ready on time | 4 |
| Total course cost | RM 0 |
| HRD Corp-claimable | RM 0 |

**Open roles**

| | |
|---|---|
| AI-Augmented Finance Analyst<br>by 24 Mar 2027 | 1 openings |
| AI-Assisted AP Specialist<br>by 24 Apr 2027 | 2 openings |
| ESG & Carbon Reporting Analyst<br>by 24 Apr 2027 | 1 openings |
| Finance Automation Specialist<br>by 24 Apr 2027 | 1 openings 1 left |

**Team**

| Person | Suggested move | Match | Weeks | Cost | On time |
|---|---|---|---|---|---|
| **Team member A (sample)**<br>Accounts Payable Clerk<br>6 h a week | AI-Assisted AP Specialist | 60% | 8 | RM 0 | Yes<br>18 Dec 2026 |
| ▸ All 4 options | | | | | |
| **Team member B (sample)**<br>Accounts Payable Clerk<br>8 h a week | ESG & Carbon Reporting Analyst | 57% | 8 | RM 0 | Yes<br>18 Dec 2026 |
| ▸ All 4 options | | | | | |
| **Team member C (sample)**<br>Accounts Payable Clerk | AI-Augmented Finance Analyst | 64% | 12 | RM 0 | Yes<br>15 Jan 2027 |

## HR redeployment planner

Who moves where, by when, at what cost. Sample team: 4 of 6 placed, 4 ready on time, RM 0 in course fees.

## Employer requirements

Employers add the skills they need; each worker’s match and route update.

## Advisor approval

A plan becomes a learning plan only after an advisor signs off.

## For every worker

Rank jobs by SDG goal, use English or Bahasa Melayu, delete your data anytime.

Live screenshot; sample employer and synthetic team, labelled in the app. The app suggests who moves where; HR makes the decision.

*SDG mapping*

# Every skill and job links to a UN goal

## One worker's route, goal by goal




Today's job

### Accounts Payable Clerk

Highly exposed to AI; the aim is redeployment, not retrenchment




A skill on the route

### GHG emissions accounting

Measures emissions so companies can cut them




Target job

### ESG & Carbon Reporting Analyst

Measures company emissions; supports sustainability disclosure

### Rank jobs by a goal

Pick a goal, such as climate action. Jobs that serve it gain 10 points on a 100-point score, and the app shows what the switch costs in match and weeks.





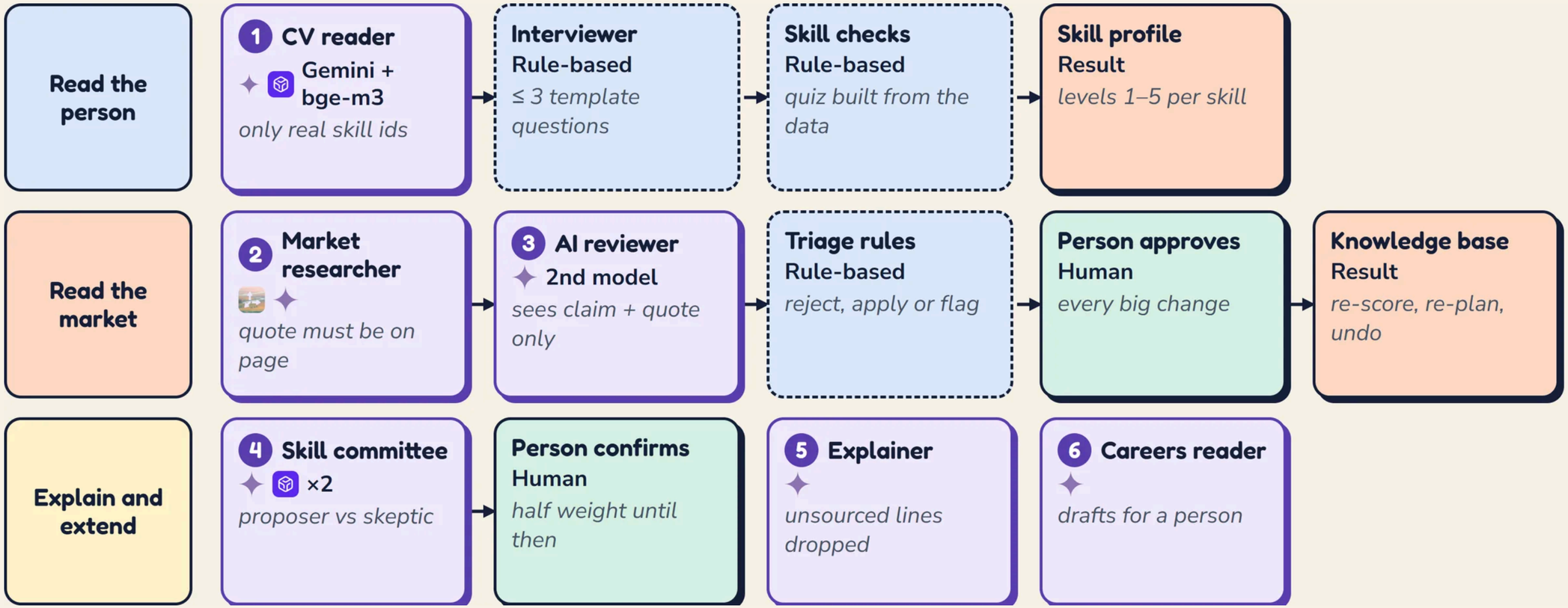


Reskilling itself serves SDG 4 (target 4.4). A test fails if any job or skill loses its tag.

Tags are our own labels on the sample catalogue, marked Sample in the app. Icons: United Nations SDGs.

*Multi-agent design*

# Six AI agents, each checked by code or a person




Purple numbered boxes are AI agents, dashed blue boxes are fixed rules, green boxes are people. Code sets every number.

*Proof*

# Every claim has a test behind it

### CV reading

**97%**

current job right on 36 CVs it never saw, with a blind answer key

### Privacy

**99.4%**

of 155 planted personal details removed before any AI reads the CV

### Exposure model

**r = 0.92**

agreement with experts on 185 jobs it never saw in training

### Independent check

**r = 0.79**

against the ILO's index on 2,956 jobs, which we never trained on

### Market researcher

**769**

articles read in one 5.5-minute live run, 7 flagged for a person

### Engine

**419**

automated tests on routes, the stress test and the HR view, all passing

Known gaps: 28% of full-ESCO skills shown are wrong (word match, wrong meaning); 1 of 155 personal details reached the AI.

*Risks*

# Our hardest risk, and the fallbacks we built

**HARDEST RISK** — **A wrong AI signal reroutes thousands of plans**

**OUR MITIGATION** — Trusted sites only. The quote must match the page word for word. A second AI reviews it. A person approves anything that changes a route, and every change can be undone.

**RISK** — **Gemini is down or out of quota**

**FALLBACK** — CV reading drops to keyword matching, labelled “basic”. The researcher replays its last run, labelled “replayed”.

**RISK** — **RunPod is cold or down**

**FALLBACK** — Embeddings switch to a local bge-m3 server or Gemini, then to keyword matching.

**RISK** — **Our AI-exposure scores are off**

**FALLBACK** — The model’s measured error is built into the 1,000-future stress test, and four sources are shown side by side.

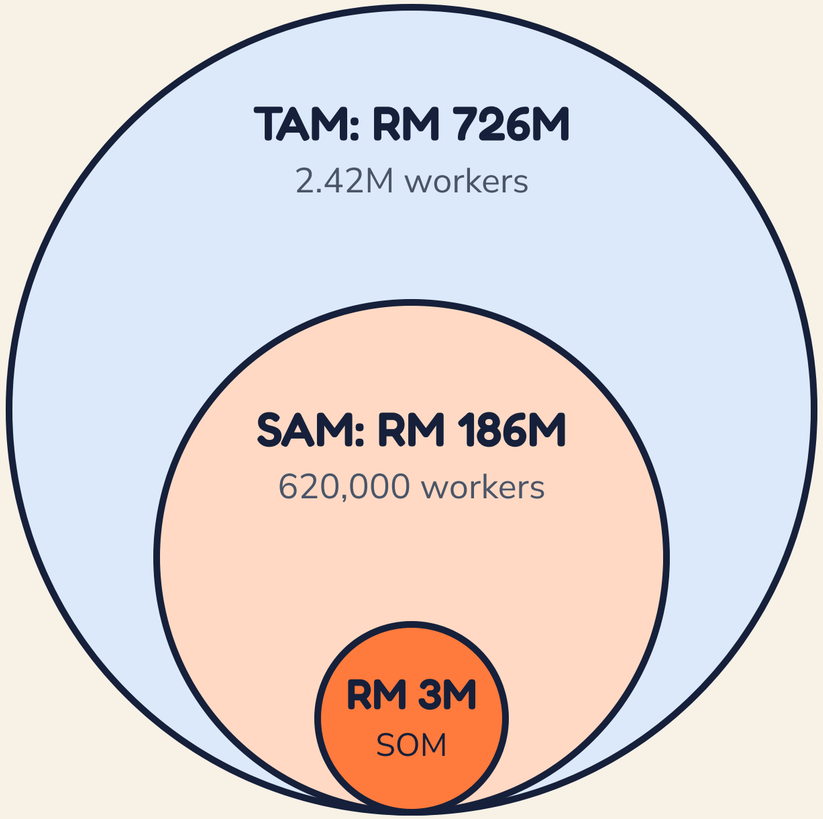
**RISK** — **Sample data taken as real**

**FALLBACK** — Every sample value is labelled Sample in the app; “check any job” runs on real ESCO data.

Every fallback is built, and the screen says when one is running.

*Market size*

# RM 726M a year in Malaysia




**TAM: workers who must move or upskill**
620k high-impact and 1.8M medium-impact jobs in 10 sectors

**SAM: jobs at high risk**
Highly impacted roles that merge or disappear within 3–5 years

**SOM: 10,000 seats by year 3**
20 employers × 500 staff, starting in Penang and KL shared-services centres (749 GBS firms)

Sources: TalentCorp impact study 2024 and GBS Malaysia Mid-Term Report 2025. All at RM 300 per worker per year, our pilot price.

*Business model*

# Free for workers, paid by employers

## Why employers pay: one clerk, 11 years' service

Statutory severance

**RM 22,800**

HalaTuju Enterprise for six months

**RM 150**

**One redeployment pays for about 150 workers.**

## Subscription models

### Enterprise — Main revenue

**RM 25 per worker a month, paid by the employer**

Redeploy staff to internal roles, or help leavers find work. HR view, and staff opt in.

### Personal

**Free**

Map, routes and stress test. Private to the worker.

### Personal Plus

**RM 9.90 a month**

Small monthly fee for certificates and advisor sessions.

### Public partners

**Annual licence**

PERKESO, TalentCorp and universities license it.

Severance: 20 days' wages per year of service at the DOSM 2025 clerical median, RM 2,693. Prices are our pilot assumptions.

*Roadmap*

# From 29 skills to 13,939, then employers

### Now: Oct 2026

Live app: 29 skills tracked daily, any of 3,039 jobs, 1,000-future stress test

**Done: live demo, 36-CV blind test**

### Finals

Track all 13,939 skills. Import HR data from employers.

**Target: letter of intent from 1 pilot employer**

### Pilot: H1 2027

Penang, home to about 90 shared-services companies such as **AMD**. First letters go to **InvestPenang** and **PSDC**.

**Target: 2–3 centres, about 1,500 staff, redeployment rate measured**

### Scale: 2027–28

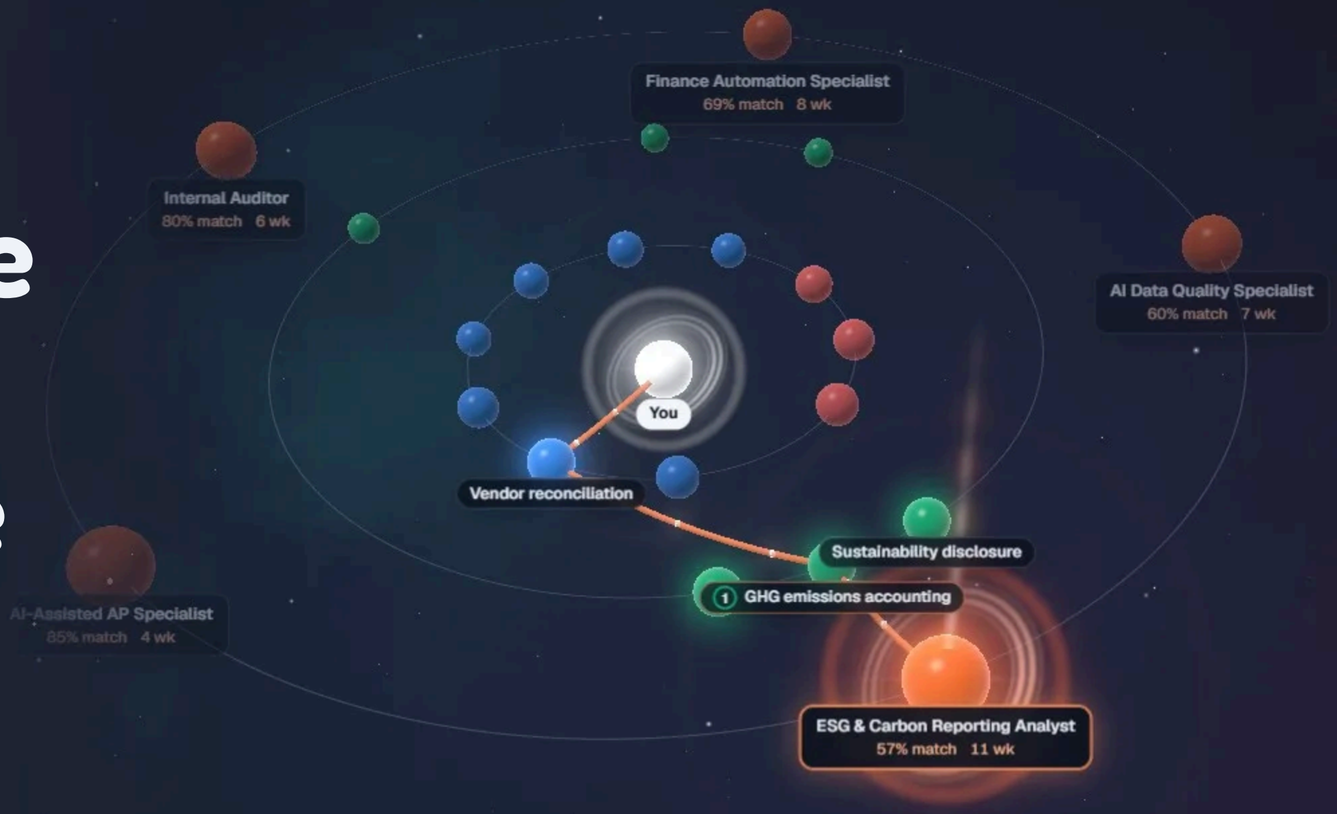
20 employers and universities. Licensed Malaysian job-ad data. More sectors.

**Target: 10,000 seats, RM 3M a year**

InvestPenang and PSDC are partners behind the Penang GBS Industry Academy. Not approached yet; AMD is an example, not a partner.

*Thank you*

# Your career, rerouted the moment AI changes the road.




## HalaTuju

Live demo: halatuju.replit.app

Code: github.com/shenenleow10521/SdcHackathon

Team Dsaise

*Appendix: CV reading*

# 36 unseen CVs, three kinds, scored blind

| Kind | CVs | Job right | Found | Correct | Privacy |
|---|---|---|---|---|---|
| Clean | 18 | 100% | 71% | 72% | 100% |
| LinkedIn | 12 | 92% | 74% | 65% | 100% |
| Messy | 6 | 100% | 68% | 77% | 98% |
| **All** | 36 | 97% | 71% | 72% | 99.4% |

On the app’s 29 map skills the same CVs score 94% found and 69% correct.

## Known gaps

- Word match, wrong meaning: “payment proposal” read as *process payments*, which ESCO means as taking payments at a till
- “SQL Accounting”, a software package, read as the SQL language in 5 CVs
- 1 of 155 personal details reached the AI, on a two-column CV

Found and correct are for the 13,939 ESCO skills. 50 synthetic CVs in all; 14 used for tuning are left out here. Median 12.9 s per CV.

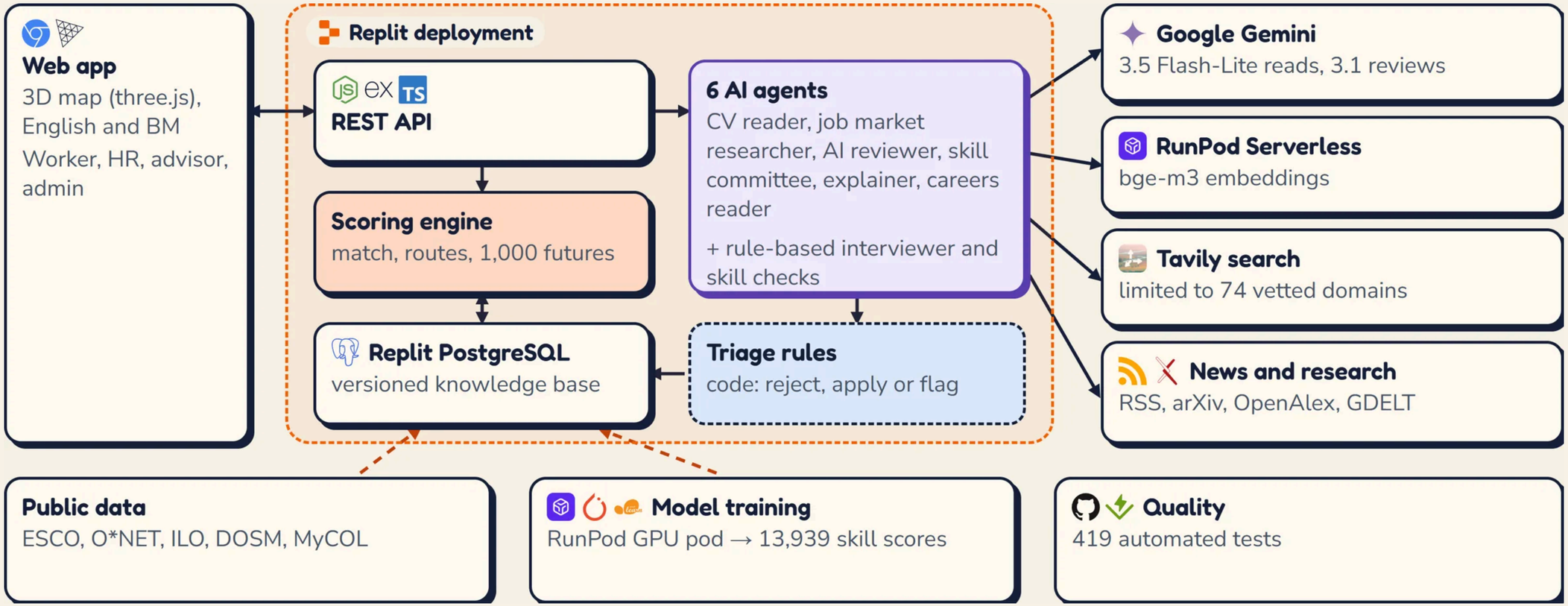
*Appendix: what sets it apart*

# Five things you can check in the live app

| What HalaTuju does | Evidence |
|---|---|
| A route to one specific job, in weeks | 11 weeks at 6 h/week, every step linked to a real course |
| Every plan stress-tested | 1,000 simulated futures (Monte Carlo); best route works in 100% |
| Reroutes from verified evidence | Quote found on the source page; a person approves |
| A measured exposure model | r = 0.92 on 185 held-out jobs; r = 0.79 vs the ILO index |
| Malaysian data on any job | DOSM wages and employment, TalentCorp critical-jobs list |

*Appendix: architecture*

# One Replit app, four outside services




AI reads text; plain code calculates every number. Dashed lines run once, offline: loading public data and training the model.

*Appendix: how it is built*

# The stack, layer by layer

## Data

**ESCO v1.2.1**
3,039 jobs and 13,939 skills

**O*NET with GPTs-are-GPTs**
16,411 expert-rated tasks

**ILO WP140**
AI exposure for 2,956 jobs

**DOSM and TalentCorp MyCOL**
Malaysian wages, critical jobs

## AI

**Gemini 3.5 Flash-Lite**
reads CVs and news

**Gemini 3.1 Flash-Lite**
reviews each claim

**bge-m3 on RunPod Serverless**
matches skills and CVs

**Exposure model on a RunPod GPU pod**
PyTorch and scikit-learn

## App

**Replit**
hosting and deployment

**Node.js, Express, TypeScript**
API and scoring engine

**Replit PostgreSQL**
versioned knowledge base

**three.js**
3D map, in English and Bahasa Melayu

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**419 automated tests pass** on GitHub with Vitest. Open source under Apache 2.0.

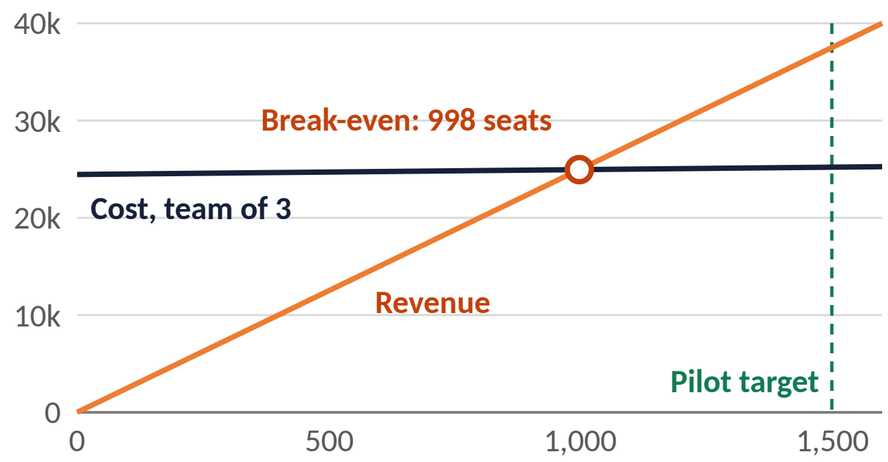
*Appendix: costs and break-even*

# Breaks even at about 1,000 workers

**Cost model, RM a month**

| | A | B | C |
|---|---|---|---|
| 1 | **Item** | **Value** | **Formula / basis** |
| 2 | Price per seat a month (RM) | 25.00 | *input* |
| 3 | Cost to serve per seat (RM) | 0.50 | *input, allowance* |
| 4 | Contribution per seat (RM) | 24.50 | =B2-B3 |
| 5 | Read one CV (RM) | 0.03 | *measured* |
| 6 | One market check (RM) | 1.05 | *measured* |
| 7 | Daily checks a month (RM) | 31.50 | =B6*30 |
| 8 | Hosting and database (RM) | 420.00 | *input, USD 100* |
| 9 | Team of 3 (RM) | 24,000.00 | *input, 3 x 8,000* |
| 10 | **Fixed cost a month (RM)** | **24,451.50** | =B7+B8+B9 |
| 11 | **Break-even seats** | **998** | =B10/B4 |
| 12 | Pilot target seats | 1,500 | *input* |
| 13 | **Profit a month at pilot (RM)** | **12,298.50** | =B12*B4-B10 |

**Revenue and cost, RM a month, by seats**




Blue = input (assumption), green = measured 24 Oct 2026, black = formula. USD 1 = RM 4.20.

*Appendix: sources*

# Every number on these slides

- ESCO v1.2.1, European Commission: occupations and skills
- ILO Working Paper 140 (2025): generative AI exposure
- Eloundou et al., “GPTs are GPTs”, Science 384 (2024)
- Anthropic Economic Index, Massenkoff & McCrory (2026)
- O*NET task statements and ratings
- TalentCorp & ILMIA, MyCOL 2024/2025
- DOSM via OpenDOSM: wages by occupation, 2021–2025
- TalentCorp AI, digital and green economy impact study (Nov 2024)
- DOSM Labour Force Survey, Sep 2025: employment by occupation
- GBS Malaysia Strategy Mid-Term Report 2022–2027 (2025)
- HRD Corp levy collection 2024 (Ministry of Human Resources)
- Employment (Termination and Lay-Off Benefits) Regulations 1980
- Microsoft Dynamics 365 blog, 22 Oct 2026 (market researcher signal)
- HalaTuju measurements, 23–24 Oct 2026: CV test, engine, tests